ROYAL EDGE — DEMONSTRATION & USER GUIDE

Rajasthan Royals IPL 2027 Auction Intelligence Platform | SupeRR Selector Season 2

1. System Architecture & Live Services

ROYAL EDGE is an enterprise-grade cricket analytics and auction intelligence platform built to identify reliable batters and bowlers under match leverage. The system runs with dual complementary interfaces: a cutting-edge React + TypeScript + Vite production web console and a standalone Streamlit contingency engine, both powered by a high-throughput FastAPI REST backend.

Service	Port / URL	Technology	Status / Operational Role
FastAPI Backend	http://127.0.0.1:8000	FastAPI / Uvicorn / Python 3.14	Core REST Engine & Statistical Computing
React Production Web App	http://127.0.0.1:5173	React 19 / TypeScript / Vite / Tailwind	Primary 8-Screen Executive Console
Streamlit Fallback Console	http://127.0.0.1:8502	Streamlit / Python Data Engine	Contingency Analytics & Rapid Prototyping

2. Walkthrough of the Eight Specialized Screens

Screen 1: Executive Overview: Presents high-level franchise macro indicators: 867 competitive matches, 200,999 delivery records, and 45,951 high-pressure balls. Provides instant preview of Top-3 BPRI Batters and Top-3 Actionable Bowlers.

Screen 2: Pressure Intelligence: Explores the ball-by-ball leverage engine. Features interactive over-by-over progression (Overs 1-20), phase concentration (Powerplay 13.2%, Middle 20.3%, Death 40.5%), and the mathematical leverage formulation P_t.

Screen 3: Batting Analytics: Houses the official audited Top 10 Batters leaderboard. Includes BPRI score, HP strike rate, HP average, death strike rate, delta vs normal play, Sample Size Saturation Index (SSSI %), and 95% bootstrap confidence intervals.

Screen 4: Bowling Analytics: Displays the official audited Top 10 Bowlers leaderboard with BOWPRI scores. Features an interactive toggle between 'BCCI-Eligible Actionable Targets' and 'All Global Benchmark Bowlers' to prevent regulatory confusion.

Screen 5: Player Comparison Studio: Enables head-to-head comparison between any two batters or bowlers across 6 normalized leverage axes on a dynamic radar chart. Automatically computes metric deltas and displays a statistical warning if 95% bootstrap confidence intervals overlap.

Screen 6: Auction War Room: Simulates live IPL auction scenarios against Rajasthan Royals' ■42.80 Cr remaining purse. Front-office users can adjust simulated bidding ceilings with real-time budget deficit alerts, squad roster tracking, and overseas slot checks (max 8).

Screen 7: Player Intelligence Profiles: Deep-dive scouting dossiers for individual candidates with phase-by-phase performance breakdowns (Powerplay, Middle, Death), empirical sample size saturation ratings, and data-backed scouting observations.

Screen 8: Methodology & Reliability: Documents the complete mathematical equations: first/second innings leverage index P_t, Bayesian shrinkage formulas (K=60 batting, K=50 bowling), composite weighting rationale, and scientific guidance on interpreting bootstrap intervals.

3. Front-Office Governance & Guardrails

1. Zero Synthetic Data: *All metrics, rankings, and indices are computed exclusively from the 200,999 deliveries in the competition dataset.*
2. Policy Segregation: *Ineligible international benchmark bowlers (e.g. Mohammad Nawaz, Abrar Ahmed) are explicitly decoupled from auction bidding pipelines.*
3. Uncertainty Transparency: *The Sample Size Saturation Index (SSSI %) represents sample accumulation relative to Bayesian stability, never future player win probabilities.*
4. Simulated Ceilings: *All bidding ceiling sliders are labeled as user-adjustable tactical simulations, not definitive player market valuations.*